

SHRAVANKUMAR NAGARAJAN

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SUMMARY

Lead Full Stack Engineer with 9+ years building scalable backend systems, RESTful microservices, and high-throughput data pipelines using Java, Python, and AWS. Experienced with PostgreSQL, Redis, observability, distributed architectures, and maintainable automated testing, with recent hands-on work building LLM-guided agent workflows, prompt guardrails, and evaluation frameworks. Currently completing an M.S. in Applied Data Intelligence with a focus on LLM, RAG, and agentic AI systems.

SKILLS

- **Languages:** Java, Python, JavaScript/TypeScript (Node.js), Scala, SQL, Bash, C/C++, Rust
- **Backend & APIs:** Spring Boot, Spring Cloud (Eureka, Config, Consul), Spring Data, Spring Reactor, FastAPI, Express.js, NestJS, REST API design, GraphQL, Apache Camel, High-Scale System Architecture, Software Testing
- **Data & Streaming:** MySQL, PostgreSQL, Oracle, DynamoDB, MongoDB, Cassandra, Redis/ElasticCache, Elasticsearch, Redshift, BigQuery, Neo4j, Kafka, Flink, PySpark, Airflow, Databricks, Trino
- **Cloud & DevOps:** AWS (ECS, EKS, S3, ElastiCache, Glue), Azure, Docker, Kubernetes, Helm, Terraform, Ansible, Jenkins, ArgoCD, Concourse CI
- **Observability:** Prometheus, Grafana, ELK Stack, OpenTelemetry, New Relic, PagerDuty, incident response
- **Frontend:** React, Next.js, Redux, micro-frontends, Tailwind CSS, Angular
- **AI / ML:** LangChain, LangGraph, Llamaindex, RAG pipelines, ChromaDB, FAISS, Hugging Face, PyTorch, scikit-learn, XGBoost, LoRA, LLM evaluation (pass@k, recall@k, precision@k), Agent Development

PROFESSIONAL EXPERIENCE

Snowflake | *Agentic AI Intern*

May 2026 - Present

- Built an agent development platform for LLM-guided RBAC automation on the Snowflake Native App runtime with Next.js, sequencing dependent objects, secrets, network rules, service accounts, and EAs into multi-step provisioning bundles for roles, grants, warehouses, databases, and network policies, enabling faster provisioning and eliminating manual configuration errors.
- Orchestrated multi-stage approval workflows across Slack, Jira, and Snowflake tasks/stages, routing requests to role and object owners and autoprovisioning on approval, cutting turnaround from days to minutes.
- Hardened LLM-generated SQL with deterministic validators and prompt guardrails, raising pass@1 from 86% to 94% and blocking 100% of unsafe DDL/DML in evaluation; validated the workflow end to end with Playwright, pytest, and Vitest.

The Bank of New York Mellon | *Lead Full Stack Developer*

Sep 2022 - Jan 2025

- Applied high-scale system architecture to design 10+ RESTful microservices in Java, Spring Boot, Spring Cloud (Eureka, Config Server, Consul), and Spring Data, improving service availability and enabling seamless scaling across AWS ECS clusters.
- Cut API p99 latency 40% under peak load through multithreading (ExecutorService, CompletableFuture), Redis/ElasticCache distributed caching, connection-pool tuning, and MySQL/PostgreSQL indexing, partitioning, and query optimization.
- Automated staged deployments on AWS EKS with ArgoCD and Helm charts, backed by parameterized Terraform, Ansible, and S3 artifact storage for IaC.
- Built React App for provisioning dashboards over a GraphQL/Express BFF, letting operations teams launch Bash, Ansible, and Terraform workflows directly from the UI.
- Owned the observability stack with Prometheus, Grafana, ELK, and PagerDuty on-call alerting, and trained XGBoost and Random Forest models for capacity forecasting and anomaly detection, reducing MTTR 45% and false alerts 35% while preventing multiple P1 outages.
- Led sprint planning for an Agile/Scrum team spanning platform, SRE, and security groups, using JIRA to coordinate work, and mentored three junior developers on microservices design patterns, which improved delivery speed and code quality

Elsevier | *Senior Software Engineer*

Jul 2020 - Sep 2022

- Built a PySpark, Groovy, and Kafka pipeline that processed 12 TB of scholarly content from S3 into Elasticsearch, powering sub-500 ms semantic search across 10M+ Scopus and Mendeley documents daily.
- Designed a star-schema Redshift warehouse with tuned distribution keys, sort keys, and materialized views, plus data-quality frameworks enforcing schema contracts and SLAs - saving \$250K annually in compute.
- Delivered document-retrieval and reviewer-recommender services in Java/Spring Boot and Python/FastAPI over MongoDB and PostgreSQL at 99.9% availability, cutting reviewer assignment time 60% through reliable backend services.
- Established software testing and performance baselines with JUnit, Testcontainers, and Gatling, isolating bottlenecks that improved Kafka consumer message processing rates 30%.

Accenture, Cognizant & Tata Consultancy Services | *Software Engineer*

May 2015 - Jul 2020

- Delivered 15+ RESTful microservices and 5+ micro-frontends for a telecom network-monitoring platform using Scala, Spring Boot, React, and Kafka, with Cassandra and DataStax Enterprise Graph consumers processing millions of topology events daily at four-nines uptime.
- Migrated a large-scale retail platform from SOAP to REST (Java/Spring), cutting p99 latency below 200 ms and doubling throughput via Oracle PL/SQL tuning and distributed caching; parallelized sequential ETL with Spring Batch, saving 10 hours per monthly cycle.
- Shipped Guidewire InsuranceNow policy and claims modules with Angular, jQuery, and Spring Boot APIs, and led a Maven-to-Gradle migration that improved build times 25% with parallel test execution.

EDUCATION

San Jose State University

Jan 2025 - Dec 2026

M.S., Applied Data Intelligence (GPA: 3.78)

San Jose, CA

- **Coursework:** Machine Learning, Deep Learning, Generative AI, Agentic AI, Big Data Algorithms, Distributed Systems, Data Warehouse & Analysis

PROJECTS

Kayak-Clone AI Travel Search & Booking Agent | *San Jose State University*

Oct 2025 - Dec 2025

- Architected a multi-agent travel concierge with LangGraph and LangChain that decomposes natural-language requests into flight, hotel, and car agent tasks, using RAG over ChromaDB embeddings and Phi-4, Mistral, Gemma 3, and OpenAI inference; achieved 92% recall and 90% precision on LangSmith evaluation benchmarks.
- Built FastAPI microservices to expose agent workflows through a React/Next.js frontend, then containerized and deployed the application with Docker on AWS.